Arivuselvan Elavarasan

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Education

K S Rangasamy College of Technology, Tiruchengode — *2025-Present*
B. Tech Artificial Intelligence & Data Science — CGPA: 9.22

Vidhya Vikas Matric Hr.Sec School — *2021-2022*
12th Grade, Tamil Nadu State Board of Higher Secondary Education, Tamil Nadu — Percentage: 95

Vidhya Vikas Matric Hr.Sec School — *2019-2020*
10th Grade, Tamil Nadu State Board of Higher Secondary Education, Tamil Nadu — Percentage: 99

Projects

Advanced Doctor Booking System

Developed a full-stack web application for seamless doctor appointment scheduling, patient management, and secure user authentication with a responsive user interface.

→ **Tools & Technologies Used:**
React, Tailwind CSS, **Java**, **Spring Boot**, Hibernate (JPA), MySQL, REST APIs
→ Designed and implemented a **responsive frontend** using React and Tailwind CSS to ensure an intuitive and user-friendly booking experience across devices.
→ Developed **backend services using Java and Spring Boot** to manage user authentication, doctor availability, appointment scheduling, and role-based access (admin/doctor/patient).
→ Implemented **RESTful APIs** for efficient communication between frontend and backend, ensuring secure and scalable data flow.
→ Integrated **Hibernate (JPA)** with MySQL for reliable data storage and retrieval, supporting patient records, appointment history, doctor profiles, and schedules.
→ Ensured **data security and validation** through Spring Security and JWT-based authentication mechanisms.

Scan & Pick

Developed a smart product identification and lookup system that allows users to scan product barcodes using a mobile device to instantly retrieve detailed product information.

→ **Tools & technologies used:** React, Node.js, Express.js, MongoDB, Barcode Lookup API, EmailJS.
→ Designed and implemented a **mobile-friendly and user-friendly interface** using React, enabling users to scan product barcodes seamlessly through their devices.
→ Integrated a **Barcode Lookup API** to fetch real-time product details such as product name, brand, category, and specifications based on the scanned barcode.
→ Developed backend services using **Node.js and Express.js** to handle API requests, data processing, and communication between the frontend and database.
→ Utilized **MongoDB** for efficient storage and retrieval of scanned product data and user interaction history.
→ Integrated **EmailJS API** to send product details, alerts, or confirmations directly to users via email after successful scans.

MACHINE LEARINING PROJECTS:

Developed an intelligent classification system using machine learning techniques to automatically analyze data and categorize inputs into predefined classes with high accuracy.

→ **Tools & technologies used:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook.
→ Implemented data preprocessing techniques including data cleaning, normalization, and feature extraction to improve model performance.
→ Trained and evaluated multiple machine learning classification algorithms such as Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM).
→ Measured model performance using evaluation metrics like accuracy, precision, recall, and F1-score.
→ Applied the trained classification model to real-world datasets for predictive analysis and automated decision-making

LAND PRICE PREDICTION:

Developed a machine learning–based predictive system to estimate land prices accurately by analyzing key factors such as location, area, amenities, and market trends.

→ **Tools & technologies used:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook.
→ Collected and preprocessed real-estate data, including handling missing values, encoding categorical features, and normalizing numerical attributes.
→ Implemented regression algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression to predict land prices.
→ Evaluated model performance using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.
→ Visualized prediction results and feature importance to support better decision-making in real estate valuation.

AI RELATED SKILLS

→ Hands-on experience using AI tools such as ChatGPT and GitHub Copilot for problem-solving and development like making prompts to get solution faster
→ Working knowledge of machine learning using Scikit-learn and TensorFlow in Jupyter/Colab
→ Working knowledge of AI-based product classification, including data preprocessing, model training, and evaluation
→ Understanding of core Machine Learning concepts including supervised learning, model evaluation, and overfitting/underfitting

Technical Skills and Interests

Front End Languages: HTML, CSS, JavaScript, ReactJS
Back End Languages: NodeJS
Developer Tools: Git, GitHub
Databases: MongoDB
Soft Skills: Communication, Adaptability
Areas of Interest: Software Development, Full-Stack Web Development,Artificial Intelligence, and Research & Development in Emerging Technologies,Machine Learning Techniques

Achievements

· Solved 100 LeetCode problems, enhancing problem-solving skills in data structures and algorithms.
· Participated in NPTEL examination and secured Elite in all the exams Participated .